

Engineering Maths

UNIT 1: CALCULUS AND LINEAR ALGEBRA

I. Limits and Continuity

- **Definition of a Limit:** The value a function approaches as the input approaches some value.
- **L'Hôpital's Rule:** Used for solving indeterminate forms like $0/0$.

II. Differentiation

- **Rules:** Chain Rule, Product Rule, and Quotient Rule.
- **Applications:** Finding the slope of a tangent line and optimization (Maxima/Minima).

III. Matrices

- **Types of Matrices:** Square, Identity, and Symmetric.
- **Inverse of a Matrix:** $A^{-1} = \frac{1}{|A|} \text{adj}(A)$.
- **Linear Equations:** Solving systems of equations using Gaussian Elimination.